

Answer any one question

1. A particle moves in one dimension according to the equation [25]

$$\ddot{x} + 9\dot{x} + 20.25x = 0,$$

with initial conditions

$$x(0) = 0, \quad \dot{x}(0) = 50.$$

Using appropriate numerical methods, answer the following. All final numerical answers must be reported correct to **5 decimal places**.

- (a) Rewrite the second-order differential equation as a system of two first-order differential equations by introducing [10]

$$v(t) = \dot{x}(t).$$

Starting from the given initial conditions, solve the system numerically over a suitable time interval (*Hint: the numbers in the equation will have to be used*) using an appropriate algorithm.

State clearly:

- i. the numerical method used,
 - ii. the step size chosen (with justification),
 - iii. the numerical values of $x(t)$ and $v(t)$ obtained.
- (b) Determine the time at which the displacement $x(t)$ is maximum. [8]
- To do this, locate the root of

$$v(t) = \dot{x}(t) = 0$$

using a suitable numerical root-finding method.

Report:

- i. the numerical method used to find the root,
 - ii. the time at which the maximum displacement occurs,
 - iii. the corresponding maximum value of $x(t)$.
- (c) Determine the minimum value of the velocity $v(t)$ for $t \geq 0$. [7]

You should use a suitable numerical procedure to identify the relevant time and then compute the corresponding velocity value.

Report:

- i. the numerical method used,
- ii. the time at which the minimum velocity occurs,
- iii. the minimum value of $v(t)$.

2. A one-dimensional random walker starts at the origin, $x = 0$. At each time step, the walker moves either one unit to the right or one unit to the left with equal probability. Thus, after each step, [25]

$$x \rightarrow x + 1 \quad \text{or} \quad x \rightarrow x - 1,$$

with probability $1/2$ each.

You are required to simulate 10^6 independent random walkers, each taking 100 steps.

For a given walker and a given position $m > 0$, define the *first passage time* T_m as the first step number at which the walker reaches the position $x = m$. If the walker does not reach $x = m$ within 100 steps, record that case appropriately in your algorithm.

Using appropriate numerical methods, answer the following.

- (a) Write an algorithm to simulate the motion of 10^6 independent one-dimensional random walkers for 100 steps each. [8]

Your algorithm should clearly indicate:

- i. how the random steps are generated,
- ii. how the position of each walker is updated at each step,
- iii. how the final position after 100 steps is stored.

- (b) For each target position [10]

$$m = 1, 2, 3, \dots, 20,$$

compute the first passage time T_m for every walker.

Using your simulation results, determine for each m :

- i. the mean first passage time,
- ii. the number (or fraction) of walkers that do *not* reach $x = m$ within 100 steps.

Present your results in a suitable table.

- (c) Plot a histogram of the final positions of the 10^6 walkers after 100 steps. [4]

Your plot should include:

- i. clearly labelled axes,
- ii. an appropriate choice of binning,
- iii. a brief numerical comment on the shape and symmetry of the histogram.

- (d) Briefly comment on how the mean first passage time changes as the target position increases from $m = 1$ to $m = 20$, based on your numerical results. [3]

3. A one-dimensional random walker starts at the origin, $x = 0$. At each time step, the walker moves either one unit to the right or one unit to the left with equal probability. Thus, after each step, [25]

$$x \rightarrow x + 1 \quad \text{or} \quad x \rightarrow x - 1,$$

with probability $1/2$ each.

You are required to simulate 10^6 independent random walkers, each taking 100 steps.

For a given walker, define the *first return time* T as the first step number at which the walker returns to the origin after leaving it. If the walker does not return to the origin within 100 steps, record that case appropriately in your program.

Using numerical simulation only, answer the following. No analytical calculation is required. You may compare your numerical results with known results if these are provided.

- (a) Write an algorithm to simulate 10^6 independent one-dimensional random walkers for 100 steps each. [8]

Your algorithm should clearly indicate:

- i. how the random steps are generated,
- ii. how the position of each walker is updated,
- iii. how the first return time is detected and stored.

- (b) Using your simulation, determine the first return time for all walkers that return to the origin within 100 steps. [8]

Hence calculate:

- i. the mean first return time,
- ii. the fraction of walkers that do not return to the origin within 100 steps.

- (c) Plot a histogram of the first return times. [6]

Your plot should include:

- i. clearly labelled axes,
- ii. a suitable choice of bins,
- iii. a brief comment on the shape of the distribution.

- (d) Briefly comment on your numerical results. In particular, state whether short return times are more common or less common than long return times, and whether returns occur at all step numbers or only at certain step numbers. [3]